

Simultaneous Group-Envelope Bounds for Γ -Robust Multiple-Choice Knapsack Problems

Anonymous submission

Abstract

Solving a Γ -robust problem through the Bertsimas–Sim threshold reduction requires bounding a nominal relaxation at each retained deviation threshold. This paper studies the cost of certifying the maximum LP value over that complete family for a robust multiple-choice knapsack problem. Exactly-one groups permit simultaneous evaluation: dualizing a fixed-threshold capacity constraint cancels threshold-dependent baselines, leaving the maximum of a constant and a common-slope line within each local deviation interval. After sorting, prefix and suffix maxima with range accumulation evaluate one multiplier at every threshold in time linear in the threshold count and nearly linear in the option count. The resulting minimax interval bound is valid by weak duality, exact on feasible singleton intervals, and no weaker than the bounded-threshold group-clique relaxation. Convex bracketing gives the deployed bound an explicit minimization certificate, so it inherits this dominance up to a prescribed tolerance. Under an otherwise identical adaptive search, the certified envelope method wins all 60 instance-level median comparisons and achieves a 2.33-fold geometric-mean speedup. Matched-interval ablations, exact epigraph LPs, and a nine-instance panel transferred from a published robust-knapsack archive support the mechanism, correctness, and out-of-generator performance. A separate exact-integer audit shows the intended boundary: the LP-certificate gain need not persist when fixed-threshold integer work dominates.

Keywords: robust optimization; multiple-choice knapsack; parametric Lagrangian bounds

Area of review: Optimization.

Subject classifications: Programming, integer: robust multiple-choice knapsack; Programming, linear: Lagrangian upper bounds; Analysis of algorithms: parametric evaluation.

1 Introduction

Cardinality-budget uncertainty reduces a robust binary problem to finitely many nominal problems indexed by a scalar protection threshold (Bertsimas and Sim, 2003, 2004). The reduction is polynomial, but a conventional method may still solve one nominal relaxation for every distinct deviation. Strong bounded-threshold formulations and divide-and-conquer reduce this burden (Álvarez-Miranda et al., 2013; Hansknecht et al., 2018; Büsing et al., 2023); the remaining task is to obtain a strong bound over an interval of thresholds without repeatedly materializing a large relaxation.

That task is practically consequential when LP relaxations, rather than the nominal integer subproblems, control runtime. In the closest large-scale study of bounded-threshold robust relaxations, Büsing et al. (2023)

report that LP solves consume 33.1% of time on average among solved instances. In their million-item robust-knapsack experiment, the share reaches 93.17%, and the root relaxation alone averages 1,062 seconds. Their model places uncertainty in the objective and therefore differs from the resource-uncertainty model below; the figures are motivation rather than transferable performance estimates. They nevertheless identify the relevant computational bottleneck directly: solving the bounded-threshold LP family can dominate a robust search.

Many operational decisions have an exactly-one menu structure: one price per product, one configuration per component, or one service level per customer segment. Every fixed-threshold relaxation of the resulting robust model is a multiple-choice knapsack linear program. Binary and at-most-one knapsacks are included by adding a dummy option. Although one fixed relaxation is classical and inexpensive, jointly bounding the full parameterized family is a different computational task.

The primary deliverable here is a faster complete threshold-disjunctive LP certificate. It can be used as a root certificate or as an interval bound in an outer threshold search for an exactly-one robust model. This is deliberately narrower than a new end-to-end integer solver: the value arises when interval LP work is material, whereas faster interval bounds cannot remove time spent inside fixed-threshold integer subproblems.

This paper identifies a cancellation created by the exactly-one equations. After dualizing a fixed-threshold capacity constraint, the threshold-dependent group baselines disappear. Within each local deviation interval, a group contributes the larger of a constant and a line whose slope is common to all active options. Prefix and suffix maxima together with range accumulation therefore evaluate one multiplier simultaneously over the full global threshold grid. The resulting algorithm is linear in the threshold count and nearly linear in the option count after sorting.

The paper makes four contributions. First, it derives the cancellation and a two-envelope algorithm that evaluates a multiplier over all thresholds at once. Second, it proves exact-minimax dominance over the closest bounded-threshold group-clique relaxation and supplies a convex bracketing procedure that transfers the theorem to the implemented bound up to an explicit tolerance. Third, it embeds the certified bound in an adaptive decomposition with global-bound invariants and a separately scoped exact-integer extension. Fourth, a fixed-design study separates algebra, multiplier certification, kernel scaling, matched-interval behavior, complete-certificate time, robustness, and external-coefficient transfer. The comparator is deliberately component-matched: the clique relaxation and the envelope bound are placed under the same search, fixed-threshold solver, stopping rule, and timing policy. The contribution is the simultaneous parametric bound, not a new uncertainty set, a faster algorithm for one knapsack relaxation, a reproduction of a complete DnC+ implementation, or universal superiority of an integer solver.

2 Robust MCKP and prior-art boundary

2.1 The Γ -robust MCKP

Let $i \in \{1, \dots, n\}$ index groups and let J_i be the finite, nonempty option set of group i . Option (i, j) has value $v_{ij} \in \mathbb{R}$, nominal resource contribution $a_{ij} \in \mathbb{R}$, and nonnegative deviation $d_{ij} \geq 0$. Exactly one

option is selected from every group. With an integer uncertainty budget $\Gamma \in \{0, \dots, n\}$, the problem is

$$\begin{aligned}
\max_x \quad & \sum_i \sum_{j \in J_i} v_{ij} x_{ij} \\
\text{s.t.} \quad & \sum_i \sum_{j \in J_i} a_{ij} x_{ij} - \max_{\substack{0 \leq u_i \leq 1 \\ \sum_i u_i \leq \Gamma}} \sum_i u_i \sum_{j \in J_i} d_{ij} x_{ij} \geq 0, \\
& \sum_{j \in J_i} x_{ij} = 1 \quad (i = 1, \dots, n), \\
& x_{ij} \in \{0, 1\}.
\end{aligned} \tag{R-MCKP}$$

For integer Γ , the inner maximum is the sum of the Γ largest selected deviations. The model covers a robust resource or residual constraint; the sign convention lets larger a_{ij} represent more nominal slack.

For a fixed binary selection, write $d_i := \sum_j d_{ij} x_{ij}$ for its selected deviation in group i . LP duality for the inner budget problem gives

$$\max_{0 \leq u \leq 1, \mathbf{1}^\top u \leq \Gamma} \sum_i d_i u_i = \min_{\theta \geq 0} \left\{ \Gamma \theta + \sum_i [d_i - \theta]_+ \right\}. \tag{1}$$

Define the global candidate set

$$\mathcal{B} := \{0\} \cup \{d_{ij} : i = 1, \dots, n, j \in J_i\} = \{b_0 < \dots < b_{B-1}\}, \quad B := |\mathcal{B}|.$$

Because the right-hand side of (1) is convex piecewise linear with breakpoints at the selected deviations, a binary selection is robust feasible if and only if there is a $\theta \in \mathcal{B}$ such that

$$\sum_i \sum_{j \in J_i} r_{ij}(\theta) x_{ij} \geq \Gamma \theta, \quad r_{ij}(\theta) := a_{ij} - [d_{ij} - \theta]_+. \tag{2}$$

Thus (R-MCKP) is a finite union of fixed-threshold MCKPs. This classical representation is the starting point.

The exactly-one form also covers binary and at-most-one selection. Append a dummy option with $(v, a, d) = (0, 0, 0)$ to an at-most-one group. More specifically, the robust binary knapsack

$$\max \left\{ \sum_i p_i y_i : \sum_i w_i y_i + \max_{0 \leq u \leq 1, \mathbf{1}^\top u \leq \Gamma} \sum_i \delta_i y_i u_i \leq W, y \in \{0, 1\}^n \right\}$$

is (R-MCKP) with two options per item: the unselected option $(0, W/n, 0)$ and the selected option $(p_i, W/n - w_i, \delta_i)$. Hence the results below include robust binary knapsack without changing asymptotic complexity.

For completeness, threshold sufficiency follows directly from convexity. For a fixed selected deviation vector q , the protection function $h(\theta) = \Gamma \theta + \sum_i [q_i - \theta]_+$ is convex and piecewise linear. Away from a selected deviation its slope is $\Gamma - |\{i : q_i > \theta\}|$, so a minimum is attained at zero or at a selected deviation, with a flat interval allowed. The global set $\mathcal{B}$ contains all such candidates.

2.2 Finite-menu pricing: MCKP derivation and use

Finite-menu pricing supplies one operational specialization. Product i has reference price $\bar{p}_i > 0$, exposure $w_i \geq 0$, and admissible prices $p_{ij} > 0$, $j \in J_i$; admissibility can encode the reference band $p_{ij} \in [(1 - \sigma_i)\bar{p}_i, (1 + \sigma_i)\bar{p}_i]$ for $\sigma_i \in [0, 1)$. Binary x_{ij} selects exactly one price, so products are the MCKP groups and candidate prices are their options. At option j , demand is $\tilde{g}_{ij} = \hat{g}_{ij} + \xi_i \delta_{ij}$, where $0 \leq \delta_{ij} \leq \hat{g}_{ij}$, $|\xi_i| \leq 1$, and $\sum_i |\xi_i| \leq \Gamma$. Let $\tilde{N}(x, \xi) := \sum_{i,j} w_i p_{ij} \tilde{g}_{ij} x_{ij}$ and $\tilde{D}(x, \xi) := \sum_{i,j} w_i \bar{p}_i \tilde{g}_{ij} x_{ij}$. Nominal revenue $\tilde{N}(x, 0)$ is maximized subject to $\tilde{N}(x, \xi)/\tilde{D}(x, \xi) \geq \Delta$ for every admissible ξ .

If $\tilde{D} > 0$ throughout, the robust ratio constraint is equivalent to requiring the worst-case residual $\tilde{N} - \Delta \tilde{D}$ to be nonnegative. Because exactly one option is selected per product, adverse signs subtract the Γ largest absolute selected residual deviations:

$$\min_{|\xi_i| \leq 1, \sum_i |\xi_i| \leq \Gamma} \{\tilde{N}(x, \xi) - \Delta \tilde{D}(x, \xi)\} = \sum_{i,j} a_{ij} x_{ij} - \max_{\substack{0 \leq u_i \leq 1 \\ \sum_i u_i \leq \Gamma}} \sum_i u_i \sum_j d_{ij} x_{ij}. \quad (3)$$

Together with nominal revenue as the objective, this is exactly (R-MCKP), with coefficients

$$v_{ij} = w_i p_{ij} \hat{g}_{ij}, \quad a_{ij} = w_i (p_{ij} - \Delta \bar{p}_i) \hat{g}_{ij}, \quad d_{ij} = |w_i (p_{ij} - \Delta \bar{p}_i) \delta_{ij}|. \quad (4)$$

Without positivity, only the additive residual interpretation remains.

To apply the paper’s results, a decision maker specifies the price menus, nominal demands, deviation scales, robustness budget Γ , and target ratio Δ , then constructs (4) once. Substitution into (2) produces one nominal MCKP relaxation per deviation threshold. The group-envelope oracle bounds these relaxations simultaneously, and the adaptive algorithm certifies the best LP value over the complete threshold family. Thus the certificate supplies a root or outer-search upper bound on robust nominal revenue; it accelerates robust price-menu optimization when interval LP work is material, but it does not estimate demand or select Γ or Δ .

2.3 Prior-art boundary

Table 1 separates the task studied here from its ingredients: threshold enumeration and filtering (Bertsimas and Sim, 2003; Álvarez-Miranda et al., 2013; Lee and Kwon, 2014), divide-and-conquer (Hansknecht et al., 2018), robust formulations and inequalities (Atamtürk, 2006; Fischetti and Monaci, 2012; Monaci et al., 2013; Joung and Park, 2021; Joung et al., 2023), and fixed-MCKP algorithms (Sinha and Zoltner, 1979; Zemel, 1980; Dyer, 1984; Szkaliczki, 2025). The closest formulation component is the bounded-threshold framework with clique aggregation of Büsing et al. (2023). None of these sources, to our knowledge, derives the baseline cancellation, two-envelope representation, certified simultaneous multiplier evaluation, or its exact-minimax dominance relation for this threshold family.

The formulation component must be distinguished from the complete algorithms in that source. DnC+ combines filtered threshold sets, strengthened estimators, optimality cuts, incumbent improvement, and early termination of nominal subproblems; its official Java implementation uses Gurobi and solves robust binary models with objective uncertainty. A direct runtime comparison would therefore change the uncertainty

placement, final deliverable, solver stack, and several algorithmic components simultaneously. Instead, the theoretical and computational comparisons here isolate the bounded-threshold clique relaxation: [Section 6](#) specializes it exactly to the group structure and evaluates it under the same adaptive search as the proposed oracle. This controlled comparison answers whether the new interval-bound component is stronger and faster; it does not reinterpret that result as a whole-solver comparison with DnC+.

The argument proceeds in three logical steps. The threshold representation first converts the robust model into a finite family of fixed-threshold linear relaxations. Lagrangian duality then replaces that family by a minimax interval certificate and permits comparison with the group-clique relaxation. Finally, the exactly-one equations expose the cancellation that makes simultaneous evaluation computationally useful. Keeping these steps separate is important: certificate validity follows from duality, whereas speed follows from the group-envelope data structure.

3 A Lagrangian bound over threshold intervals

Fix $\theta \in \mathcal{B}$ and define the group baseline

$$r_i^*(\theta) := \max_{j \in J_i} r_{ij}(\theta), \quad c_{ij}(\theta) := r_i^*(\theta) - r_{ij}(\theta) \geq 0, \quad C(\theta) := \sum_i r_i^*(\theta) - \Gamma\theta.$$

Using the exactly-one equations, (2) is equivalent to $\sum_{i,j} c_{ij}(\theta)x_{ij} \leq C(\theta)$. If $C(\theta) < 0$, the fixed-threshold problem is infeasible. If $C(\theta) \geq 0$, choosing a maximizer of $r_{ij}(\theta)$ in every group gives zero total transformed cost, so the disjunct is feasible. Its LP relaxation value is

$$L(\theta) := \max \left\{ \sum_{i,j} v_{ij}x_{ij} : \sum_{i,j} c_{ij}(\theta)x_{ij} \leq C(\theta), \quad \sum_j x_{ij} = 1, x \geq 0 \right\}. \quad (5)$$

Define the feasible-threshold index set $\mathcal{F} := \{k \in \{0, \dots, B-1\} : C(b_k) \geq 0\}$ and assume $\mathcal{F} \neq \emptyset$; if it is empty, every threshold disjunct is infeasible. The target certificate is

$$M := \max_{k \in \mathcal{F}} L(b_k), \quad (6)$$

an upper bound on the integer robust optimum obtained by relaxing every disjunct. More explicitly, if Z_k^{IP} is the integer optimum of threshold disjunct k , then the classical union representation gives $Z^{\text{robust}} = \max_{k \in \mathcal{F}} Z_k^{\text{IP}} \leq \max_{k \in \mathcal{F}} L(b_k) = M$.

Dualize the capacity constraint of (5) with $\lambda \geq 0$:

$$D(\lambda, \theta) := \lambda C(\theta) + \sum_i \max_{j \in J_i} \{v_{ij} - \lambda c_{ij}(\theta)\}. \quad (7)$$

For a contiguous index interval $I = \{\ell, \ell+1, \dots, u\} \subseteq \{0, \dots, B-1\}$, set

$$U(I) := \inf_{\lambda \geq 0} \max_{k \in I \cap \mathcal{F}} D(\lambda, b_k), \quad (8)$$

where $\max \emptyset := -\infty$. If interval I has a nonempty finite evaluated multiplier set $\Lambda_I \subseteq [0, \infty)$, define $\widehat{U}_{\Lambda_I}(I) := \min_{\lambda \in \Lambda_I} \max_{k \in I \cap \mathcal{F}} D(\lambda, b_k)$.

Proposition 1 (Validity, finite-search safety, and singleton exactness). *For every interval I ,*

$$\max_{k \in I \cap \mathcal{F}} L(b_k) \leq U(I) \leq \widehat{U}_{\Lambda_I}(I).$$

For a feasible singleton $I = \{k\}$, $U(\{k\}) = L(b_k)$.

Proof. If $I \cap \mathcal{F} = \emptyset$, then every displayed quantity equals $-\infty$ by convention, so the claim is immediate. Assume henceforth that $I \cap \mathcal{F} \neq \emptyset$. For every $k \in \mathcal{F}$ and $\lambda \geq 0$, weak duality gives $L(b_k) \leq D(\lambda, b_k)$. Maximizing over $k \in I \cap \mathcal{F}$ and then taking the infimum over λ gives the first inequality. Restricting an infimum to Λ_I can only increase its value, giving the second. On a feasible singleton, (7) is the Lagrangian dual obtained after optimizing over the product of group simplices. The feasible bounded LP (5) satisfies strong duality, hence $L(b_k) = \inf_{\lambda \geq 0} D(\lambda, b_k)$. $\square$

To compare the minimax bound formally with the closest prior-art relaxation, let $\underline{\theta} = b_\ell$, $\bar{\theta} = b_u$, and define $Q(I)$ as the optimum of the bounded-threshold group-clique LP

$$\begin{aligned} \max_{x, p, z} \quad & \sum_{i, j} v_{ij} x_{ij} \\ \text{s.t.} \quad & \sum_{i, j} \left[-a_{ij} + [d_{ij} - \bar{\theta}]_+ \right] x_{ij} + \sum_i p_i + \Gamma z \leq -\Gamma \underline{\theta}, \\ & \sum_j [\min\{d_{ij}, \bar{\theta}\} - \underline{\theta}]_+ x_{ij} \leq p_i + z & (i = 1, \dots, n), \\ & \sum_j x_{ij} = 1 & (i = 1, \dots, n), \\ & x, p \geq 0, \quad 0 \leq z \leq \bar{\theta} - \underline{\theta}. \end{aligned} \tag{9}$$

This is the exactly-one specialization of the bounded-threshold clique construction of Büsing et al. (2023). We use the extended-value convention $Q(I) = -\infty$ if this maximization problem is infeasible.

Theorem 2 (Dominance by the exact minimax envelope). *For every threshold interval I , the exact minimax group-envelope bound is no larger than the bounded-threshold group-clique bound:*

$$\max_{k \in I \cap \mathcal{F}} L(b_k) \leq U(I) \leq Q(I).$$

Thus $U(I)$ is at least as strong as the comparator relaxation for the objective under study, while avoiding an interval LP solve.

Proof. If $I \cap \mathcal{F} = \emptyset$, then $U(I) = -\infty$ and both inequalities are immediate. Hence assume $I \cap \mathcal{F} \neq \emptyset$. The epigraph formulation of (8) is a linear program. Its dual has variables $\mu_k \geq 0$ and $y_{kij} \geq 0$ and can be

written

$$\begin{aligned}
\max \quad & \sum_{k \in I \cap \mathcal{F}} \sum_{i,j} v_{ij} y_{kij} \\
\text{s.t.} \quad & \sum_{k \in I \cap \mathcal{F}} \mu_k = 1, \quad \sum_j y_{kij} = \mu_k \quad (k, i), \\
& \sum_{k,i,j} c_{ij}(b_k) y_{kij} \leq \sum_k \mu_k C(b_k).
\end{aligned}$$

Strong duality applies because the primal epigraph is feasible and has a finite optimum. The dual can be read as a convex mixture of threshold-indexed group-simplex points coupled by one aggregate capacity inequality. Take any feasible dual solution and set

$$\begin{aligned}
x_{ij} &:= \sum_k y_{kij}, & z &:= \sum_k \mu_k (b_k - \underline{\theta}), \\
p_i &:= \sum_{k,j} [\min\{d_{ij}, \bar{\theta}\} - b_k]_+ y_{kij}.
\end{aligned}$$

The simplex equations and bounds on z follow immediately. For every $b_k \in [\underline{\theta}, \bar{\theta}]$,

$$[\min\{d_{ij}, \bar{\theta}\} - \underline{\theta}]_+ \leq (b_k - \underline{\theta}) + [\min\{d_{ij}, \bar{\theta}\} - b_k]_+,$$

so the group-clique rows hold after multiplication by y_{kij} and summation. Expanding the aggregate capacity row of the dual and using $\sum_j y_{kij} = \mu_k$ cancels the group baselines and gives

$$\sum_{k,i,j} r_{ij}(b_k) y_{kij} \geq \Gamma \sum_k \mu_k b_k.$$

The identity $[d - b_k]_+ = [d - \bar{\theta}]_+ + [\min\{d, \bar{\theta}\} - b_k]_+$ then gives the first constraint of (9). The mapped point has objective $\sum_{k,i,j} v_{ij} y_{kij}$, so every feasible dual value is attained by a feasible clique-LP point. Maximizing and applying strong duality proves $U(I) \leq Q(I)$. $\square$

The finite evaluated bound $\widehat{U}_{\Lambda_I}(I)$ remains valid by [Proposition 1](#), but an arbitrary coarse multiplier set need not inherit [Theorem 2](#). [Theorem 6](#) below closes this gap by quantifying the distance between an explicitly evaluated bound and $U(I)$.

The order of maximum and infimum matters. Generally $\max_k \inf_{\lambda} D(\lambda, b_k) \leq \inf_{\lambda} \max_k D(\lambda, b_k)$, so a nonsingleton interval bound can exceed the complete fixed-threshold scan. Successive splitting tightens this relaxation, and complete refinement to feasible singletons eliminates it; the evaluation algorithm below removes its computational bottleneck.

4 Simultaneous group-envelope evaluation

Theorem 3 (Baseline cancellation). *For every $\lambda \geq 0$ and $\theta \in \mathcal{B}$,*

$$D(\lambda, \theta) = \sum_i \max_{j \in J_i} \{v_{ij} + \lambda r_{ij}(\theta)\} - \lambda \Gamma \theta. \quad (10)$$

Proof. Substitute $C(\theta) = \sum_i r_i^*(\theta) - \Gamma \theta$ and $c_{ij}(\theta) = r_i^*(\theta) - r_{ij}(\theta)$ into (7). For each group,

$$\lambda r_i^*(\theta) + \max_j \{v_{ij} - \lambda[r_i^*(\theta) - r_{ij}(\theta)]\} = \max_j \{v_{ij} + \lambda r_{ij}(\theta)\}.$$

Summing and retaining $-\lambda \Gamma \theta$ proves the identity. $\square$

The cancellation is consequential because the remaining threshold dependence has only two forms. For a group i , define

$$A_i(\lambda, \theta) := \max_{j: d_{ij} \leq \theta} \{v_{ij} + \lambda a_{ij}\}, \quad (11)$$

$$Q_i(\lambda, \theta) := \max_{j: d_{ij} > \theta} \{v_{ij} + \lambda(a_{ij} - d_{ij})\}, \quad (12)$$

where the maximum of an empty set is $-\infty$.

Lemma 4 (Two-envelope representation). *For $\lambda > 0$,*

$$\max_j \{v_{ij} + \lambda r_{ij}(\theta)\} = \max\{A_i(\lambda, \theta), Q_i(\lambda, \theta) + \lambda \theta\}. \quad (13)$$

Between consecutive distinct deviations of group i , A_i and Q_i are constant. Hence the group contribution is the maximum of one constant and one line of slope λ on that local interval, with at most one crossover.

Proof. If $d_{ij} \leq \theta$, then $r_{ij}(\theta) = a_{ij}$ and the option score is constant in θ . If $d_{ij} > \theta$, then $r_{ij}(\theta) = a_{ij} - d_{ij} + \theta$, and the score has intercept $v_{ij} + \lambda(a_{ij} - d_{ij})$ and slope λ . Maximizing separately over the two sets yields (13). Those sets are unchanged between consecutive local deviations. $\square$

4.1 Prefix-suffix range algorithm

Sort each group by d_{ij} . Before multiplier queries, applying the same unmasked accumulation with $v_{ij} = 0$ and $\lambda = 1$ computes every $C(b_k)$ and hence $\mathcal{F}$. The global threshold indices split into maximal contiguous blocks on which the local saturated set $\{j : d_{ij} \leq b_k\}$ is unchanged; all options tied at a block's left endpoint are saturated. At a fixed λ , prefix maxima of $v_{ij} + \lambda a_{ij}$ supply the constant values in (11), while reverse suffix maxima of $v_{ij} + \lambda(a_{ij} - d_{ij})$ supply the active-line intercepts in (12). If both envelopes are finite on a block, their crossover is $(A_i - Q_i)/\lambda$; a binary search finds the first global threshold at or above it. If one envelope is empty, the other applies on the entire block. Two difference arrays accumulate the resulting constant and linear pieces; cumulative sums recover the group total at all global thresholds. Finally subtract $\lambda \Gamma b_k$.

Algorithm 1 One simultaneous multiplier evaluation

Require: Global thresholds $b_0 < \dots < b_{B-1}$; deviation-sorted groups; feasibility mask $\mathcal{F}$; $\lambda \geq 0$

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1: if  $\lambda = 0$  then
2:   Set every entry to  $\sum_i \max_j v_{ij}$ 
3:   Mask indices outside  $\mathcal{F}$  by  $-\infty$ ; return the vector
4: end if
5: Initialize intercept and slope difference arrays of length  $B + 1$ 
6: for each group  $i$  do
7:   Compute prefix maxima of  $v_{ij} + \lambda a_{ij}$ 
8:   Compute suffix maxima of  $v_{ij} + \lambda(a_{ij} - d_{ij})$ 
9:   for each local deviation interval do
10:    Read its constant  $A$  and active-line intercept  $Q$ 
11:    if  $A = -\infty$  then
12:      Range-add  $Q + \lambda b_k$  on the interval
13:    else if  $Q = -\infty$  then
14:      Range-add  $A$  on the interval
15:    else
16:      Locate the clipped first index with  $b_k \geq (A - Q)/\lambda$ 
17:      Range-add  $A$  before that index and  $Q + \lambda b_k$  from it onward
18:    end if
19:  end for
20: end for
21: Cumulatively sum both arrays and subtract  $\lambda \Gamma b_k$ 
22: Mask indices outside  $\mathcal{F}$ ; return the  $B$  values
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Theorem 5 (Exact evaluation and complexity). *Let $K := \sum_i |J_i|$. In real arithmetic, after sorting the global thresholds and each group, Algorithm 1 returns the values in (10) on $\mathcal{F}$ and $-\infty$ elsewhere for a fixed multiplier in*

$$O(B + K \log(B + 1)) \text{ time and } O(B + K) \text{ working storage.}$$

Constructing the global threshold set and sorting the groups costs $O(K \log K)$. A direct score evaluation at every option–threshold pair takes $\Theta(BK)$ arithmetic operations; fully materializing those scores also takes $\Theta(BK)$ storage.

Proof. Lemma 4 proves exactness on every local interval, including repeated deviations and endpoints under the convention $d_{ij} \leq \theta$ for the saturated set. Each group produces at most $|J_i| + 1$ local intervals. Prefix and suffix maxima cost $O(K)$. Every local interval performs constant work and one binary search in the B global thresholds, totaling $O(K \log(B + 1))$. Moreover, setting all $v_{ij} = 0$ and $\lambda = 1$ in (10) yields $\sum_i \max_j r_{ij}(b_k) - \Gamma b_k = C(b_k)$, so the same range algorithm constructs the complete feasibility mask within the stated complexity. The cumulative sums and mask application cost $O(B)$. The sorted group arrays, local-interval records, and two difference arrays occupy $O(K + B)$ storage. $\square$

4.2 Certified minimization over the multiplier

Write $\phi_I(\lambda) := \max_{k \in I \cap \mathcal{F}} D(\lambda, b_k)$ and define

$$H_I := \sum_i \max_{j \in J_i} \max\{|a_{ij}|, |a_{ij} - d_{ij}|\} + \Gamma \max_{k \in I} |b_k|. \quad (14)$$

The implementation brackets a minimizer before contracting the bracket. Starting from zero and a positive scale, it doubles the positive endpoint while the evaluated objective strictly decreases. Once two consecutive values are nondecreasing, the interval from zero to the latter point contains a minimizer. Golden-section comparisons then contract this interval. The algorithm returns the smallest explicitly evaluated value at a multiplier inside the final bracket; it does not use an optimizer-interpolated objective.

Theorem 6 (Certified deployed minimax bound). *For every interval with $I \cap \mathcal{F} \neq \emptyset$, ϕ_I is finite, convex, piecewise linear, and H_I -Lipschitz on $[0, \infty)$. The geometric expansion above terminates after finitely many evaluations and produces a bracket $[\ell, u]$ containing a minimizer. For every evaluated $\bar{\lambda} \in [\ell, u]$,*

$$U(I) \leq \bar{U}(I) := \phi_I(\bar{\lambda}) \leq U(I) + H_I(u - \ell) \leq Q(I) + H_I(u - \ell). \quad (15)$$

Consequently, stopping when $H_I(u - \ell) \leq \eta_I$ yields a valid evaluated interval bound whose excess over the exact minimax value, and over the clique-dominance benchmark, is at most η_I . For a feasible singleton, $L(b_k) \leq \bar{U}(\{k\}) \leq L(b_k) + \eta_{\{k\}}$.

Proof. For fixed k , (10) is a sum of maxima of affine functions of λ , plus an affine term. It is therefore finite, convex, and piecewise linear. A slope of any active affine piece is $\sum_i r_{ij_i}(b_k) - \Gamma b_k$. Because $r_{ij}(b_k)$ lies between $a_{ij} - d_{ij}$ and a_{ij} , its absolute value is bounded by H_I . A finite pointwise maximum preserves convexity, piecewise linearity, and this common Lipschitz bound, proving the first claim.

In the baseline form (7), $c_{ij}(b_k) \geq 0$ and every group contains an option with zero transformed cost. Thus each $D(\cdot, b_k)$ has eventual slope $C(b_k) \geq 0$ for $k \in \mathcal{F}$. The finite maximum ϕ_I is consequently non-decreasing after its last breakpoint. Geometric expansion therefore reaches consecutive endpoints $a < b$ with $\phi_I(b) \geq \phi_I(a)$. Convexity then implies that some minimizer lies in $[0, b]$; otherwise all subgradients through b would be negative, contradicting the nonnegative secant slope. Standard golden-section comparisons preserve at least one minimizer in the retained bracket.

Finally, if λ^* is a minimizer in $[\ell, u]$, Lipschitz continuity gives $0 \leq \phi_I(\bar{\lambda}) - \phi_I(\lambda^*) \leq H_I|\bar{\lambda} - \lambda^*| \leq H_I(u - \ell)$. Substitute $\phi_I(\lambda^*) = U(I)$ and apply Theorem 2; singleton exactness follows from Proposition 1. $\square$

After bracketing, golden-section contraction needs $O(\log(H_I(u - \ell)/\eta_I))$ simultaneous evaluations. The released solver uses target $\eta_I \leq 10^{-9} + 10^{-8} \max\{1, |\bar{U}(I)|\}$. Its padded rectangular group arrays require $O(B + nm_{\max} + K)$ storage, reducing to $O(B + K)$ in the equal-menu experiments. The returned evaluated value is advanced by one binary64 representable number toward $+\infty$, while the corresponding lower certificate is rounded once toward $-\infty$; the reported gap includes both steps. The theorem concerns real arithmetic; binary64 feasibility and termination use stated tolerances, so the implementation is a solver-tolerance certificate rather than an interval-arithmetic proof.

5 Adaptive certification

Let E be the set of thresholds whose fixed LPs have been solved, initialized so that $E \cap \mathcal{F} \neq \emptyset$; the feasibility mask supplies such an index whenever $\mathcal{F} \neq \emptyset$. Let $\mathcal{P}$ be a collection of active intervals covering every index

in $\mathcal{F} \setminus E$; its intervals may also contain evaluated indices. Let U^{disc} be the maximum last valid bound of every removed interval not yet dominated by the incumbent, with $U^{\text{disc}} = -\infty$ initially. Maintain

$$\text{LB} := \max_{k \in E \cap \mathcal{F}} L(b_k), \quad \text{UB} := \max \left\{ \text{LB}, U^{\text{disc}}, \max_{I \in \mathcal{P}} \bar{U}(I) \right\},$$

where a maximum over an empty collection is $-\infty$. The algorithm begins with the root interval, evaluates its endpoints and midpoint, and repeatedly bisects the active interval with largest bound. The same oracle-independent endpoint–midpoint choices are evaluated in every new nonsingleton interval for both compared methods. If the three root candidates contain no feasible threshold, the initialization requirement above calls for one additional index from the already constructed feasibility mask before any gap test. All timed comparison instances had a feasible endpoint or midpoint, so this fallback was not invoked there. An interval can be removed once its stored bound is within the requested tolerance of LB; its last valid upper bound is retained in U^{disc} until LB dominates it. Singleton thresholds are solved directly.

Proposition 7 (Certificate invariant). *At every iteration, $\text{LB} \leq M \leq \text{UB}$. If the procedure stops when*

$$\text{UB} - \text{LB} \leq \varepsilon \max\{1, |\text{LB}|\},$$

then it certifies (6) to scaled absolute–relative tolerance ε . If all feasible thresholds have been evaluated, then $\text{LB} = M$ and all interval records may be removed, setting $\text{UB} := \text{LB}$. Alternatively, if every active and retained discarded bound is already at most LB, then the maintained bounds satisfy $\text{LB} = \text{UB} = M$.

Proof. LB is the maximum over a subset of fixed-threshold LP values. Every unevaluated feasible threshold belongs to an active interval or to a removed interval whose retained bound is represented by U^{disc} ; [Proposition 1](#) therefore implies $M \leq \text{UB}$. Splitting replaces a parent by children whose union is the parent, and direct singleton evaluation transfers its exact LP value into LB. These operations preserve the inequalities. The stopping claim follows by division by $\max\{1, |\text{LB}|\}$. If every feasible threshold has been evaluated, the definition of M gives $M = \text{LB}$ and obsolete interval records can be deleted. Under the alternative bound-dominance condition, the displayed definition of UB gives $\text{UB} = \text{LB}$; in either case the stated equality follows. $\square$

The primary certificate concerns M , the complete threshold-disjunctive LP upper bound. The released implementation also provides an exact integer variant: it replaces the LP lower bound by a globally robust integer incumbent, solves a fixed-threshold MCKP branch-and-bound whenever a singleton must be resolved, and prunes a whole interval whenever its retained LP upper bound cannot improve the incumbent.

Corollary 8 (Exact integer integration). *The integer interval-search variant maintains $Z^{\text{incumbent}} \leq Z^{\text{robust}} \leq \text{UB}$. If it exhausts the active intervals or terminates with $\text{UB} - Z^{\text{incumbent}}$ within the prescribed scaled tolerance, the incumbent is globally optimal to that tolerance.*

Proof. Every fixed-threshold integer optimum is bounded by its fixed-threshold LP value, and [Proposition 1](#) bounds all such LP values in an active interval. Direct singleton solution transfers an exact integer value into the incumbent; splitting preserves interval coverage; and pruning occurs only when a valid retained upper

bound cannot improve the incumbent. The invariant and stopping claim follow exactly as in [Proposition 7](#). $\square$

6 Computational study

6.1 Design and comparators

The certified-minimization protocol was serialized before the final rerun; its SHA-256 digest is 6db8550e9e9bb47f352bac93423d82067ed09e6ba6cbe4608a7544c2e3a7d261. The factorial instance design predates the certified oracle, and the complete final protocol is serialized with the release; it is a reproducible fixed design rather than an external preregistration. All runs used one thread on an Apple M4 under macOS, Python 3.14.2, NumPy 2.4.4, and SciPy 1.17.1 with HiGHS. Source data, raw timing repetitions, environments, summaries, and generation scripts accompany the paper.

Four structured families isolate distinct difficulties: *dense frontier* gives many competitive value–resource tradeoffs; *correlated risk* couples value gains and deviations; *near tie* produces unstable rankings; and *many breakpoints* gives essentially unique deviations, so B grows with K . The primary panel uses six options per group, $\Gamma = \lfloor \sqrt{n} \rfloor$, $n \in \{360, 720, 1440\}$, five seeds per family–size cell, and five timing repetitions per method. Method order alternates within an instance; the instance median is the sole statistical observation. We report geometric-mean paired speedup and a 10,000-draw design-stratified bootstrap interval that resamples seeds within each family–size cell. These descriptive summaries quantify variation across the fixed generated design; they do not define a superpopulation of robust MCKPs.

The complete-certificate comparator is the bounded-threshold LP of [Büsing et al. \(2023\)](#), specialized with one clique/GUB inequality per exactly-one group as in (9) and assembled in sparse CSR form. Both methods use the same best-first splitting policy, fixed-threshold LP routine, endpoint–midpoint candidate rule, scaled tolerance $\varepsilon = 10^{-6}$, and time limit including preprocessing. They differ only in the interval upper bound: the comparator solves a bounded-threshold clique LP, whereas the compressed method evaluates the group-envelope Lagrangian bound.

This is an end-to-end comparison for the stated deliverable—certifying the maximum LP value over the complete threshold family—and a component isolation relative to the broader algorithms of [Büsing et al. \(2023\)](#). It is not a reconstruction of DnC+: importing that solver would add threshold filtering, estimators, cuts, incumbent operations, early termination, and a different commercial solver stack. Holding the surrounding search fixed makes the source of any bound, work-count, or timing difference identifiable.

The shared fixed-threshold routine constructs each group’s upper value–cost hull and greedily merges its slopes, avoiding branch-and-bound data that neither certificate needs. Held-out validation compares this LP-only path with an independent reference routine at every threshold; both certificates therefore use the same fixed-MCKP solver.

Protocol-fixed gates required maximum algebraic error at most 2×10^{-6} ; kernel speedup at least $3\times$ for $n \geq 360$; 100% primary tolerance completion; geometric-mean primary speedup at least $2\times$ with interval lower endpoint above $1.5\times$; at least 80% timing wins; root dominance in at least 95% of cases; a median win in all four families; and a median win in every robustness configuration.

6.2 Correctness validation and oracle ablations

Forty held-out irregular instances use unequal menu sizes, repeated and near-repeated deviations, signed objectives, and Γ values from 0 to n . We compare the compressed trace with (i) the dense baseline-cost trace, (ii) direct evaluation of (10), (iii) independently optimized interval bounds, (iv) exact epigraph LP solutions for randomly selected intervals, (v) an independent fixed-MCKP LP routine at every threshold, and (vi) a complete threshold scan. The maximum algebraic discrepancy is 5.59×10^{-8} ; the maximum violation of the certified upper/lower enclosure against the exact epigraph LP is 0.00; and the largest scaled minimization certificate is 1.00×10^{-8} . The minimum root-bound slack over the complete scan is 0.00. All serialized validation gates pass.

The kernel ablation separates algebraic compression from adaptive search. It contains 48 instances, nine normalized multiplier queries, five repetitions, and balanced order. Table 2 reports geometric-mean query and total construction-plus-query speedups. The maximum identity error is 9.78×10^{-9} . For $n \geq 360$, total speedup is 21.05 $\times$ in geometric mean. In the many-breakpoints family, where B grows linearly with K , the descriptive log-log slopes are 0.94 for the compressed evaluator and 2.00 for dense materialization; the left panel of Figure 1 shows the separation. Storage denotes persistent numerical-array bytes rather than peak process memory and grows sharply only in that family.

To isolate the interval oracles from adaptive-tree effects, a common-trace ablation evaluates both bounds on the same 168 dyadic intervals from 24 independently generated instances. The envelope oracle wins 24 instance timings with a 2.53 $\times$ geometric-mean speedup; every evaluated bound is certified, with maximum scaled minimization gap 9.56×10^{-9} , and its bound is no larger than the clique bound on 100.0% of matched intervals. This comparison includes preprocessing and sparse LP assembly. Because the intervals are identical, it is the direct isolation test of the interval-bound component; the adaptive experiment below then measures the additional effect of tighter bounds on the search trajectory.

6.3 Primary result

Both methods reach tolerance on all 60 primary instances. Every envelope bound is certified, with maximum scaled minimization gap 9.43×10^{-9} . The envelope method wins all 60 paired timings, with geometric-mean speedup 2.33 $\times$ and design-stratified 95% interval [2.16, 2.51]. It evaluates a median 28.8% of thresholds as fixed LPs. Final lower bounds are identical at recorded precision, and its root bound never exceeds the clique root bound within numerical tolerance. At the raw repeat-block level, 298/300 are wins (minimum speedup 0.79 $\times$); median within-instance coefficients of variation are 0.60% and 0.39%. Across an instance’s interval solves, the clique comparator uses a median 100 394 sparse-matrix nonzeros and 1.19 MiB of cumulative CSR arrays. Together with the matched-interval result, these observations show that the gain survives complete certification and is not produced by different fixed-threshold solvers or search policies. The right panel of Figure 1 displays instance- and size-level scaling.

Every family median exceeds one, from 1.34 $\times$ for many breakpoints to 6.30 $\times$ for dense frontier, so the protocol-fixed breadth gate passes.

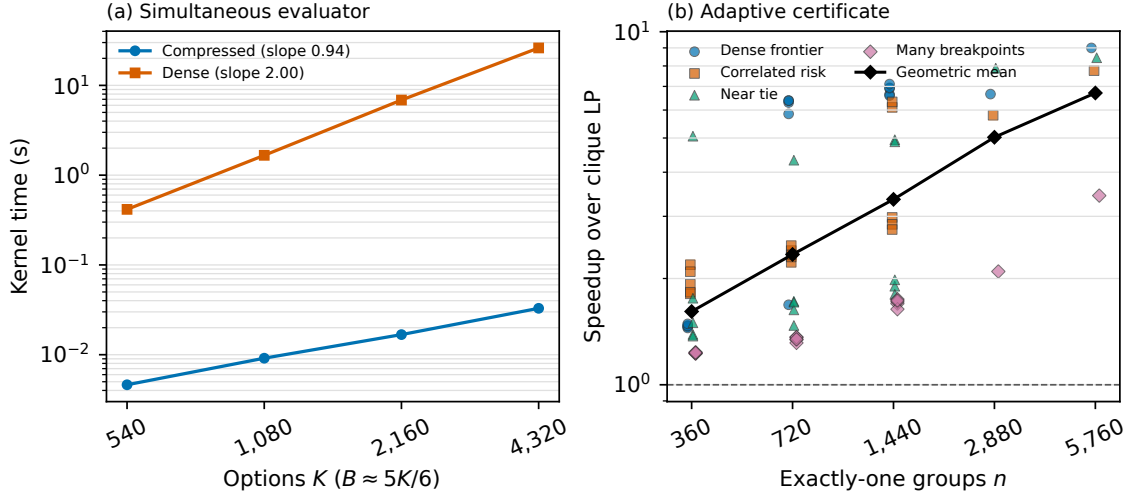


Figure 1. Mechanism and end-to-end scaling. Left: construction plus nine multiplier queries in the many-breakpoints kernel panel, where B grows linearly with K ; lines show geometric means over three seeds, with descriptive log–log slopes. Right: instance-level speedup of the compressed adaptive certificate over the clique-LP certificate; black diamonds are geometric means by size. Primary observations use five seeds at $n \leq 1440$ and stress observations use one separate seed at $n \in \{2880, 5760\}$.

6.4 Robustness, stress, and published-coefficient evidence

The 36-instance robustness panel changes one dimension at $n = 720$: $\Gamma = 1$, twelve-option menus, or $\Gamma = \lfloor 0.2n \rfloor$. Every instance reaches tolerance, and the compressed method wins all 36 timings; every configuration-level bootstrap interval remains above one. The larger fraction of fixed thresholds under the large budget partly explains its lower speedup than the wide-menu panel.

The descriptive stress panel uses four families, one new seed, and two repetitions at each of 2,880 and 5,760 groups. All eight instances reach tolerance with identical final lower bounds. The geometric-mean speedup is $5.80\times$ overall and $6.71\times$ at 5,760 groups. The many-breakpoints instances contain 14,401 and 28,801 thresholds, respectively; detailed rows are in the companion.

For an out-of-generator check, we use the published robust-knapsack archive of [Gersing et al. \(2022\)](#). Each binary item becomes the two-option group given after (R-MCKP). We retain the archive’s nominal profits, weights, budgets, and deviation vectors, but transfer its objective deviations to resource deviations; this is a transparent model-compatible coefficient stress test, not a reproduction of the source uncertain-objective model. The panel contains three published seeds at each of 1,000, 5,000, and 10,000 items. Both methods reach tolerance, every envelope bound is certified with scaled gap at most 9.85×10^{-9} , and the method wins 9/9 timings with geometric-mean speedup $2.31\times$, rising to $3.07\times$ at 10,000 items.

6.5 Exact integer integration audit

The integer integration in [Corollary 8](#) was audited separately from the LP-certificate experiment. Twelve instances use 30 groups, all four families, three new seeds, two balanced timing repetitions, and a five-second limit. The envelope and clique interval searches use the same fixed-threshold branch-and-bound and cer-

tify 11 of 12 instances each; complete threshold enumeration certifies all 12, while compact SCIP certifies 11. Whenever at least two methods certify, their objectives differ by at most 1.14×10^{-13} . On the 11 jointly certified envelope–clique cases, the geometric-mean clique/envelope time ratio is 0.96, showing that fixed-threshold integer search, rather than interval-bound computation, dominates at this scale. The many-breakpoints family remains the decisive interval-search limitation: enumeration and compact SCIP certify all three instances while each interval method certifies two.

This result is a bottleneck decomposition rather than counterevidence to the LP-certificate result. The envelope oracle reduces the interval-bound component; it cannot accelerate fixed-threshold branch-and-bound once that component controls total time. The audit therefore validates the exact integration and locates its operating boundary without supporting universal integer-solver superiority. The demonstrated computational claim remains acceleration of the complete LP-bound certificate, particularly in the root and outer-search regime where bounded-threshold relaxations are material.

6.6 Application-derived semi-synthetic panel

The pricing model in [Section 2.2](#) also supplies an application-derived realism check. We use the UCI Online Retail data set, which contains 541,909 transaction records ([Chen, 2015](#)). The released calibration reads the first 200,000 records; after positive-price/quantity filtering and the eight-observation SKU rule, 192,451 observations contribute to 2,549 SKU aggregates. These aggregates calibrate the segment mix, price levels, volumes, and uncertainty scales of nine semi-synthetic pricing portfolios; documented elasticity and menu-generation assumptions remain modeled rather than observed. For each portfolio, the reported certificate is an upper bound on the best robust-feasible nominal revenue across the complete price-risk threshold family. The panel uses 360, 720, and 1,440 groups, twelve price choices, three new seeds per size, and three balanced repetitions. Both methods reach tolerance, and the compressed method wins 9/9 instances with geometric-mean speedup $1.46\times$ (95% cell-stratified bootstrap interval $[1.24, 1.75]$), rising to $1.70\times$ at 1,440 groups. This panel tests scale and coefficient realism; demand estimation is outside the algorithmic scope, and the nonrandom row cap and modeled demand curves preclude causal or transaction-level validation claims.

6.7 Scope and limitations

The experiments validate the envelope identity, certified minimax approximation, retained LP certificate, and speed advantage over the tested sparse clique-LP implementation, not universal integer-solver dominance. The public-data panel transfers coefficients rather than reproducing the source uncertain-objective problem, and the pricing panel remains semi-synthetic. The implementation uses Python/SciPy on one architecture, while compiled code could shift constants; streaming could reduce dense-oracle storage without changing its $\Theta(BK)$ arithmetic. As established in the introduction and comparator design, the official implementation of [Büsing et al. \(2023\)](#) targets objective uncertainty and combines several algorithmic components with Java/Gurobi. The controlled comparison therefore isolates its closest bounded-threshold clique relaxation rather than claiming a direct reproduction of DnC+.

The exact audit shows that fixed-threshold integer search can erase the interval-oracle advantage and

that compact SCIP remains preferable on many-breakpoints instances. The study directly supports complete LP-family certification and interval bounding in the tested adaptive search. Repeated node-level bounds, pricing-loop re-solves, and sensitivity sweeps are plausible integration settings but are not evaluated here. DnC+, specialized fixed-MCKP solvers, and the oracle are therefore complementary.

7 Conclusion

For exactly-one robust models, the proposed oracle makes the complete threshold-disjunctive LP certificate a more practical root or outer-search bound. After sorting, group envelopes evaluate one multiplier over all thresholds in $O(B + K \log(B + 1))$ time and $O(B + K)$ working storage. Convex bracketing converts those evaluations into an explicit minimization certificate: the deployed bound is valid, within a prescribed tolerance of the exact minimax value, singleton-exact to that tolerance, and no larger than the clique bound plus that tolerance.

The evidence is both component-matched and end-to-end for this deliverable. On identical interval traces the proposed oracle is no weaker throughout and faster on every instance; under the shared adaptive search, the study records zero certificate violations, 60 of 60 primary wins, a $2.33\times$ geometric-mean speedup, and 9 of 9 wins on published knapsack coefficients. The exact integration locates the boundary: integer subproblem work can dominate, and compact SCIP remains stronger on many-breakpoints instances. Integration into full robust-solver stacks is therefore the natural computational extension, while multiple robust constraints remain the clearest theoretical extension.

Data and code availability

The anonymous supplement contains source code, tests, the protocol and digest, instance generators, raw timing repetitions, result tables, environment records, and the script that regenerates every numerical artifact.

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Table 1. Boundary between the present task and related algorithmic streams.

Stream	Established operation	Treatment here
Bertsimas–Sim and filtered enumeration	Solve a finite, possibly reduced set of nominal threshold problems	Simultaneously evaluates a Lagrangian multiplier over all thresholds
Fast fixed MCKP LP	Solves one MCKP relaxation using hull or partition structure	Amortizes evaluation across a parameterized family; does not improve the one-LP complexity
Divide-and-conquer	Avoids threshold solves using monotonicity	Replaces an interval LP by a direct exactly-one group-envelope bound
Strong and bounded-threshold formulations	Strengthen or linearize the robust model over a threshold interval	Proves objective-bound dominance over the group-clique interval LP without introducing a new robust formulation
Ellipsoidal robust MMKP (Caserta and Voß, 2019)	Treats multiple resources and covariance-based uncertainty	Different uncertainty set, geometry, and computational target

Table 2. Kernel ablation against dense threshold materialization. Speedups are geometric means over four families and three seeds. Storage is the median (maximum) ratio of persistent numerical-array bytes and excludes interpreter overhead.

Groups	Cases	Query speedup	Total speedup	Storage ratio	Maximum error
90	12	5.94	11.22	2.58 (35.3)	1.7e-10
180	12	8.56	14.85	2.60 (70.5)	5.8e-10
360	12	11.82	19.05	2.61 (140.9)	2.3e-09
720	12	14.90	23.26	2.62 (281.6)	9.8e-09

Table 3. Primary complete-certificate comparison. Each reported time and evaluation count is the median across instances of that size after taking the within-instance median over five balanced timing repetitions. Aggregate speedup treats each instance once. In the fixed-threshold evaluation column, C denotes the compressed certificate and Q the clique certificate.

Groups	Cases	Compressed (s)	Clique (s)	Geometric speedup	Wins	Fixed-threshold LP evaluations (C / Q)	Clique interval LP evaluations
360	20	0.186	0.308	1.61	20/20	12/12	8
720	20	0.349	0.703	2.34	20/20	12/12	8
1440	20	0.660	1.666	3.35	20/20	12/12	8

Table 4. Robustness configurations at 720 groups, four families, and three new seeds. Within each configuration, bootstrap intervals resample the three seeds separately by family.

Configuration	Geometric speedup [95% interval]	Wins	Thresholds evaluated
Sparse budget ($m = 6, \Gamma = 1$)	1.82 [1.62, 2.27]	12/12	11.5%
Wide menu ($m = 12, \Gamma = \lfloor \sqrt{n} \rfloor$)	2.81 [2.24, 3.52]	12/12	28.8%
Large budget ($m = 6, \Gamma = 0.2n$)	2.77 [2.56, 3.10]	12/12	34.6%